Michealmas Week 7 & 8

lw703

1 Completed Task

We did experiment on multi-step DiKL in toy task: MOG-3 and it works well. This dataset consists of a three-component 2D Gaussian mixture with a location scaling of 15 and a log-variance scaling of 0.01, which serves as the target distribution. To validate the method, we employ a two-step sampling procedure, ensuring that the hyperparameters remain consistent with those used in the single-step neural sampler.

In particular, an additional intermediate time step $t = 0.5$ is introduced between time steps 1 (representing the data distribution) and 0 (denoting noise). The Gaussian kernels are defined as:

$$\tilde{k}_{0.5}(\tilde{x}|x) = \mathcal{N}(0.5x, \sqrt{0.75}), \quad k_{0.5}(x|x) = \mathcal{N}(0.2x, 0.82I), \tag{1}$$

$$\tilde{k}_{0,1}(\tilde{x}|x) = \mathcal{N}(0.1x, \sqrt{0.99}I). \tag{2}$$

The training of the two-step model is performed using the following optimization steps:

$$\theta_1 = \arg\min_{\theta_1} \mathrm{DiKL}\left(f_{t_1}^{\theta_1}(x) * k_{0.5} \middle\| p_d * \tilde{k}_{0,1}\right), \tag{3}$$

$$\theta_2 = \arg\min_{\theta_2} \mathrm{DiKL}\left(f_{t_2}^{\theta_2}(f_{t_1}^{\theta_1}(x)) * \tilde{k}_{0.5} \middle\| p_d * \tilde{k}_{0.5}\right). \tag{4}$$

The results, as illustrated in Figure 3, indicate that our method significantly mitigates the issue of mode-seeking.

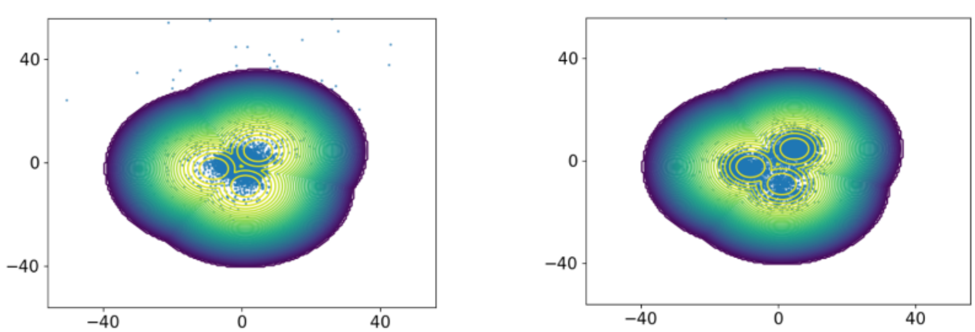

Figure 1: Results of the two-step sampling process.

2 Future Work

An alternative method to increase the model capacity is to add the post-processing Langevin dynamics to get final distribution $\tilde{p}_{\theta,\lambda}(x)$. While the neural sampler trained using the diffusive KL (DiKL) divergence achieves high-quality samples, its ability to fully capture complex distributions remains constrained. Future work can integrate Bayesian learning [**?**] to dynamically adapt these hyperparameters, improving convergence and efficiency.